Inferential statistics

STC224

Inferential statistics ::

Association - Bivariate correlation […]

THE OUTPUT

- The Pearson correlation (Pearson r) measures both the strength of the correlation and the direction of the correlation. The value runs from 0 to 1.
- Any observed r of greater than 0.5 or -0.5 indicates strong positive relationship and strong negative relationship respectively.
- Strong positive means when one variable increases, the other increase as well (uphill regression line). Negative means when one variable increases, the other decreases (downhill regression line).
- The Sig. (2-tailed) indicates whether we can accept of reject the null hypothesis that says the two variables are **not** related.
- Correlations best go together with Scatter plots we saw in the last chapter for adding visual.

CHAPTER 7 ::

INFERENTIAL STATISTICS - MEAN DIFFERENCES

IN THIS TOPIC

- T-Test
    - One Sample T-Test
    - Independent Samples T-Test
    - Paired Samples T-Test

Inferential statistics :: Mean differences – T-test

- Inferential statistics of mean difference compare two means, expected and observed to see if they are significantly different.
- The resulting outcome tells whether a treatment (independent variable) significantly affects the dependent variable and/or the extent to which the resulting outcome (dependent variable) can be explained by the independent variable.

Inferential statistics ::

Mean differences – One Sample T-test

- One sample T-test compares a sample mean (observed) to a known population (expected) mean to see if they significantly differ than would have happened by chance.
- For example, if we know the average birth-weight of babies in rural areas, then we select 100 mothers for an intervention to increase it. We will test if the intervention successfully improved birth-weight by comparing the known average value to the observed value after the intervention.
- The null hypothesis is that the observed and expected means do not differ.

Inferential statistics ::

Mean differences – One Sample T-test […]

- Select “analyze” □ “Compare means” □ “One sample T-test”
- Move the variable under observation to the “Test Variables” box.
- Enter the known population mean “1000” into the “Test value” box. Click [Ok]

Inferential statistics ::
Mean differences – Independent Samples T-test

- The independent samples T-test compares the means of two groups of a sample created by a dichotomous categorical variable (e.g. married, not married) to see if they differ significantly. If they do, then we say that the explaining (independent) variable affects the outcome (dependent variable).
- In our example, we will differentiate the “Family Income” means of people who said they saw a dance performance and those who said they didn’t. This will help us explain whether those who said they didn’t might have done so because they are poor.
- Our null hypothesis is that the two means are not significantly different, thus family income does not affect whether they attended a dance performance.

Inferential statistics ::

Mean differences – Independent Samples T-test […]

- Go to “Analyze” □ “Mean differences” □ “Independent-Samples T-test”
- Move the variable to be explained (the scale variable) “Family income” to the “Test Variable” box. Move the dichotomous variable “Attended dance performance last year” into the “Grouping variable” box.
- Click “Define groups” to tell SPSS what values you were using for the groups. In this case enter 0 and 1 and click [continue]. Click [Ok]

Inferential statistics ::

Mean differences – Independent Samples T-test […]

- Under Levene’s Test of Equal Variances, if equal variances assumed has Sig less that .05 (=> Homogeneity of variances can’t be assumed), use the next row (equal variances not assumed). If not, then use the first row.
- Can we reject the null hypothesis?

Inferential statistics ::

Mean differences – Paired Samples T-test

- The paired samples T-test compares the means of two variables created by repeated measurements (pre-test, post-test) of the same sample.
- For example the initial scores of students after using a lecture method of teaching, with the scores of the same students after using a student-centered approach.
- For our example though, we will compare the initial state of cancer patients and their state after 2 weeks of treatment.
- What would be our null hypothesis? What will the result tell us?

Inferential statistics ::

Mean differences – Paired Samples T-test […]

- Go to “Analyze” □ “Compare means” □ “Paired-samples T test”
- Select both “TOTALCIN” and “TOTALCW2” at once from the variables list to the selection using the transfer arrow. Click [Ok]
- Looking at the Sig., what can we conclude?

CHAPTER 8 ::
Inferential Tests – Non-Parametric Tests

- While parametric tests (Tests of association and Mean differences) discussed in the last 2 Chapters used parameters or assumptions like Normality and Linearity (e.g. equal variance or equal means), Non-parametric tests do not require such.
- Thus they just deal with differences in frequencies.
- Non-parametric tests are good at analyzing categorical variables. Thus, while we were using scale variables as dependent or variables to be explained in the previous tests, non-parametric tests will use categorical variables to be explained.
- In other words parametric tests like T-tests, ANOVA, Correlations and Regressions require mathematical manipulation of continuous data (e.g. means and Std. deviations). But we cannot calculate these on categorical variables, so non-parametric tests that depend on only frequency counts can be used.

Inferential Tests ::

Non-parametric tests – Chi-square of Independence

- The Chi-Squared Test of Independence is a test of correlation between two or more nominal variables.
- We have done the procedure before when we were producing Cross-tabs. For a recap, the following slide is what we did with cross-tabulation in Chapter 6 :: Inferential Statistics of Association.

{Recap}

Inferential statistics :: Association - Cross-tabs

- The cross-tabulation procedure works well with two categorical variables provided that one of them is a dichotomous (e.g. Yes, no) variable. The responses are cross-sectioned to see how respondents belonging to one group responded to another question.
- Ware going to see how the people who said they were married or not rated their self happiness. Our null hypothesis is "*marital status is* **not** *related to self happiness*".
- Go to "Analyze", "Descriptive statistics", "Crosstabs".
- Transfer "Married" to the "rows" selection box and "Self rated happiness" to the "columns" selection box.
- Under [Statistics] check "chi-square" as our inferential test and click [continue].
- Under [cells], on the "percentages" box, check "rows". Click [continue] then [Ok].

Inferential Tests ::

Non-parametric tests – Chi-square of Independence […]

SECOND EXAMPLE

- We are going to see the relationship between sex and program type.
- What is our null hypothesis?
- Go to “Analyze” □ “Descriptive Statistics” □ “Cross-tabs”.
- Transfer “Sex” to the “Rows” box and “Program Type” to the “Columns” box.
- Check to produce clustered bar charts.
- Under [Statistics], choose “Chi-Square”
- Under [Cells] select “Observed” and “Expected” under “Counts”; and “Row”, “Column”, “Total” under “Percentages”.
- Click [Ok]

Inferential Tests ::

Non-parametric tests – Chi-square of Goodness-of-Fit

- The Chi-square Goodness-of-fit test has a null hypothesis that assumes that the groups created by a categorical variable are equal. We will reject this null hypothesis if the groups are significantly different at the 0.05 confidence interval.
- We are going to see if the 3 major subjects (Neuroscience, Biology and Psychology) are equally attractive to students.
- Our null hypothesis is that they are equally attractive or there is no significant difference between them.

Inferential Tests ::

Non-parametric tests – Chi-square of Goodness-of-Fit […]

- Go to “Analyze” “Non-parametric tests” “One Sample”.
- Go to the “Fields” Tab”. Remove all variables from the “Test Fields” box by transferring them to the “Fields” box except “Major”.
- Click [Run].
- You may double-click the output to view the “Model Viewer” which outlines the details for the whole procedure.